H2 ANDERSON JUNIOR COLLEGE
HIGHER 2
ANSWERS

2018 JC2 PRELIMINARY EXAMINATIONS

CANDIDATE NAME

PDG

INDEX NUMBER

H2 BIOLOGY 9744/02

Paper 2 Structured Questions

11 SEPT 2018
TUESDAY

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graph
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/9
2	/9
3	/11
4	/11
5	/11
6	/10
7	/11
8	/9
9	/11
10	/8
	/100

Answer **all** questions

1 Fig. 1.1 shows the effect of pH on the activity of a protease enzyme at the optimal temperature of 37°C.

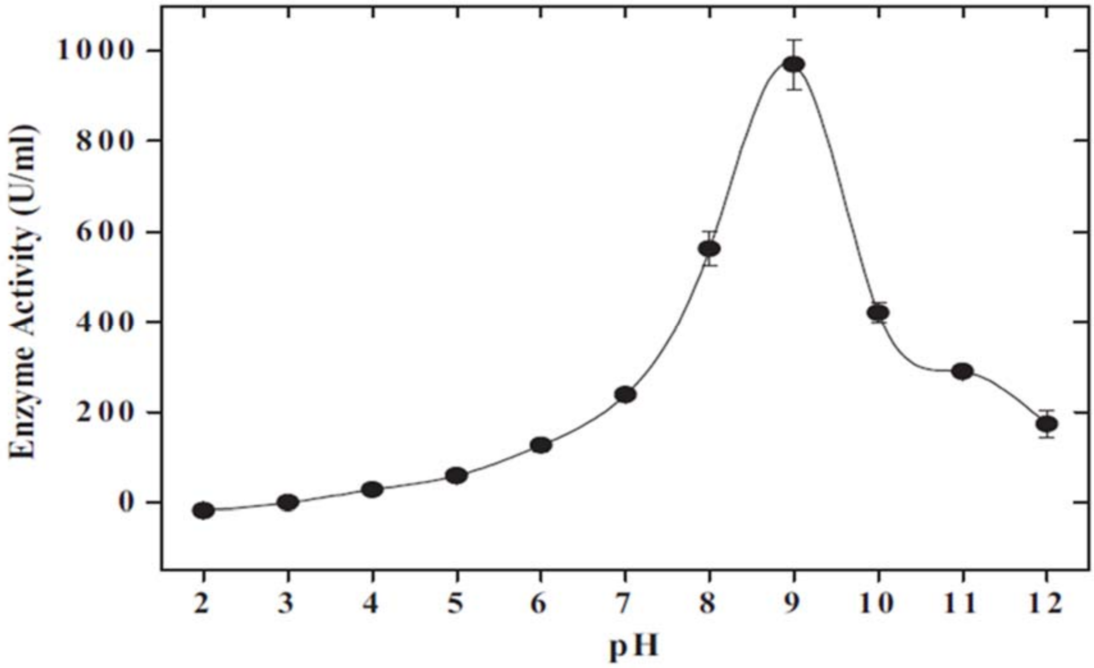


Fig. 1.1

(a) Draw, on Fig. 1.1, the approximate shape of the curve if the same experiment is conducted at 25°C.

- Similar shape to Fig. 1.1 but lower than curve in Fig. 1.1; [1]

(b) Explain with reasons the shape of the curve you have drawn.

- At lower temperature, lower kinetic energy of molecules;
- Less effective collisions between enzyme and substrates, less enzyme-substrate complex formation per unit time / lower rate of enzyme-substrate complex formation; [2]

(c) Using information from the graph, explain why proteases stored in vesicles with pH 7.2 cannot break down vesicular membrane proteins and suggest how these proteases can be activated through increase in pH.

(At least one)

- With low activity of just 200 U/ml, enzyme is inactive at pH 7.2;
- Optimal enzyme activity of 900 U/ml (accept reasonable figure quoted, correct units quoted) is highest at pH 9;

At pH 9 (accept argument at pH 7.2):

- charges on acidic and basic R-groups altered;
- Contact and catalytic residues has the correct charge to catalyse the reaction at pH 9 (accept converse);
- R-group interactions such as ionic and hydrogen bonds are altered;
- Tertiary structure / 3D conformation / configuration of enzyme is that of an active enzyme;
- Active site shape is complementary to shape of substrate;
- Maximum rate of enzyme-substrate complex formation;

[max 4m]

(At least one)

- pH can be increased to optimum pH by pumping of H^+ ions out of the vesicles
- Vesicle membrane has proton pumps; [6]

[Total: 9]

2 RNA molecules play important roles within cells. One of the major types of RNA found in all cells is the ribosomal RNA (rRNA).

Fig. 2.1 shows rRNA molecules forming the large ribosomal subunit in eukaryotes.

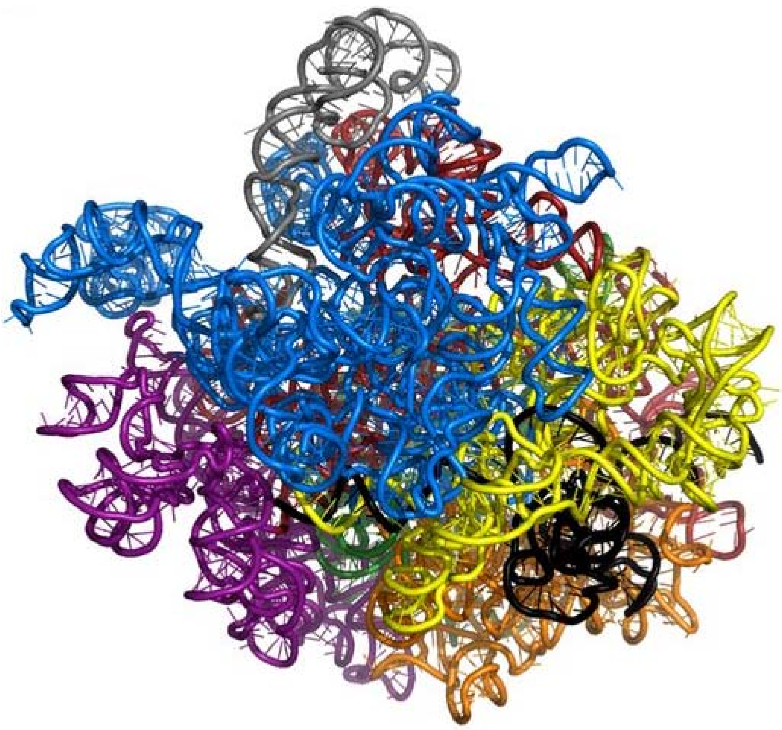


Fig. 2.1

(a) Explain why the rRNA molecules must adopt the shapes shown in Fig. 2.1.

How each shape is formed (max 2)

- **Single stranded RNA folds back** upon itself due to
- **Complementary base-pairing** via **hydrogen bonds** / presence of **complementary** stretches of **bases** that can form **hydrogen bonds** (
- Forming a **double helical** structure

Purpose

- to **stabilize the molecule**

- *Max 2 marks for binding*: to form a **binding site** whose **shape is complementary** to **ribosomal proteins / small ribosomal subunit / tRNA molecules**

[3]

(b) Another important RNA molecule found in eukaryotic cells is the telomerase RNA. Telomerase RNA is found within the telomerase enzyme, an enzyme essential for elongating telomeres.

(i) Outline how RNA molecules such as telomerase RNA and rRNA are synthesised.

- **Transcription** of
- the **gene** for **telomerase RNA/ rRNA**
- **by RNA polymerase**
- catalyzing formation of **phosphodiester bonds**
- between **RNA nucleotides**
- which were added via **complementary base pairing**
- using (one of the two strand of) **DNA** as **template**
- elongation occurs in the **5' to 3'** direction of the RNA strand

(½ m each) [3]

Fig. 2.2 shows the mode of action of telomerase.

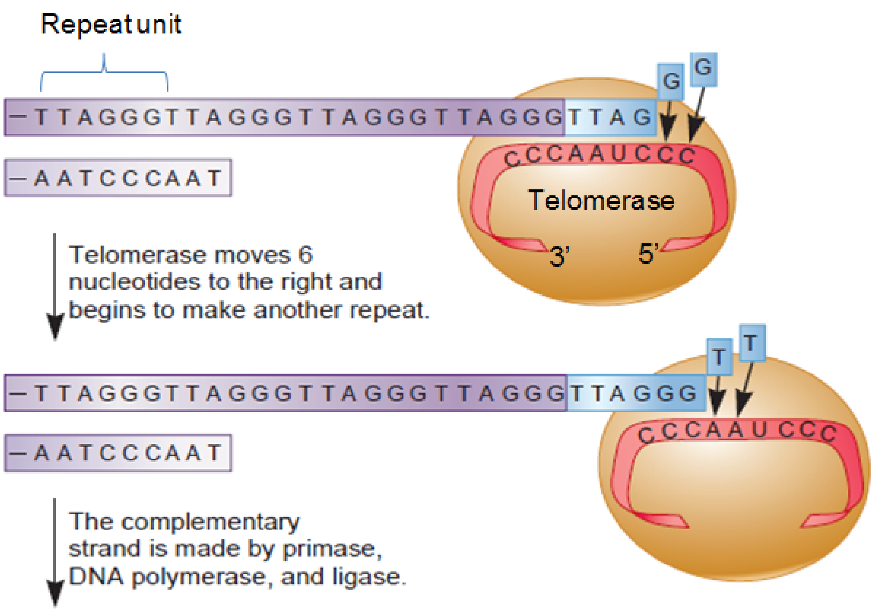


Fig. 2.2

(ii) Describe **three** visible differences between telomere elongation shown in Fig. 2.2 and translation.

	Telomere elongation	Translation
Monomers	**DNA nucleotides**	**Amino acid**
Bonds formed between monomers	**Phosphodiester bonds**	**Peptide bonds**
Enzyme involved in synthesis the bond between monomers	**Telomerase/ reverse transcriptase**	**Peptidyl transferase**
Movement of enzyme	Moves a distance of **6 nucleotides** each time	Moves a distance of **3 nucleotides / 1 codon** each time

[3]

[Total:9]

3 Eukaryotes regulate the expression of their genes at various levels of protein synthesis.

(a) Describe the effect of histone acetylation on gene expression.

Any 2

- when **lysine** on histones tails are acetylated their **positive charges** are **neutralized**
- there is **decreased interactions of histone tails** with **neighbouring nucleosomes / decreased interaction of histone tails with the (negatively charged) DNA / DNA less tightly wound around histones**
- **Chromatin** structure become **less compact/ condensed /** forms **euchromatin**

Compulsory pt– to answer " effect on effect on gene expression)

- Transcriptional machinery like **RNA polymerase** *(reject DNA polymerase)* and **transcription factors** *(must give these two egs)* will can **bind** to the **promoter of genes** in the acetylated region to / form the **transcription inititation complex**/ Allow transcription to occur / gene to be transcriptional active

[3]

(b) Eukaryotic gene expression can also be regulated at translation initiation after the mRNA is synthesised.

Fig. 3.1 shows translation occurring on an eukaryotic mRNA.

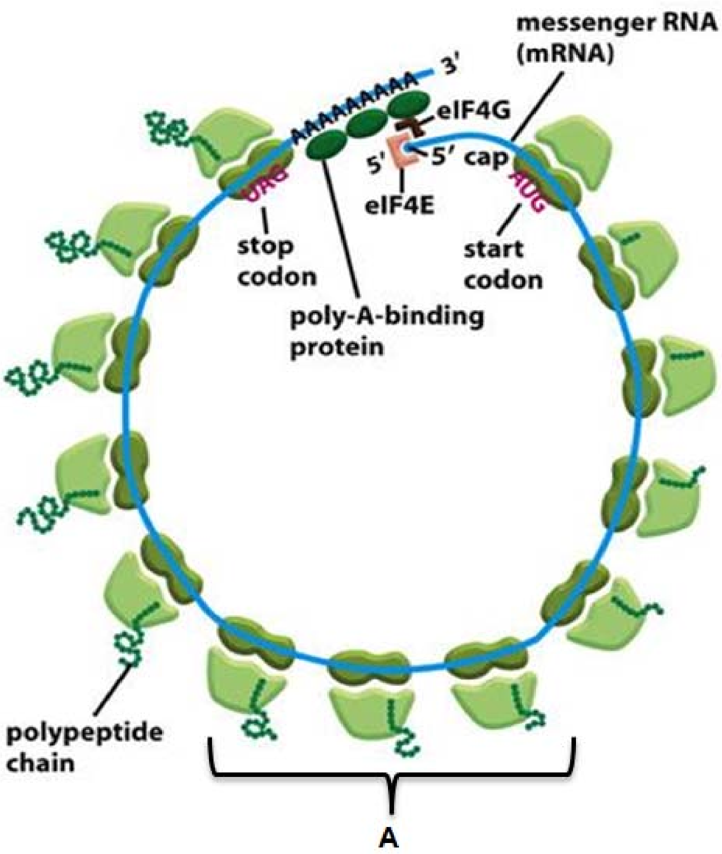


Fig. 3.1

(i) Explain the significance of the pattern of translation, labelled **A** in Fig. 3.1**.**

- Identify A: **Polyribosomes / polysomes/ multiple ribosomes** work on translating **one** messenger RNA

- Significance of A**: single** mature mRNA is used to make **many copies of the same / identical polypeptide simultaneously /** enables a cell to synthesise **many copies of the same type of polypeptide very quickly**

- **single** mature mRNA is used to make **many copies** of the **same (type)** of **polypeptide simultaneously /** enables a cell to synthesise **many copies** of the **same type of polypeptide very quickly**

[2]

(ii) During translation initiation, translation initiation factors like eIF4E and eIF4G form part of a complex which aid in recruiting ribosomal subunits to mRNA.

With reference to Fig. 3.1, describe the role of the poly-A tail and 5' cap in the assembly of ribosomes.

- **poly-A binding protein** binds to the **poly-A tail**
- which in turn recruit/ attract **eIF4G** to bind to **poly-A binding proteins**
- **eIF4E** binds to **5'cap**
- **eIF4E binds** to **eIF4G**
- to assemble into(part of) **translation initiation complex** (which aid in recruiting ribosomal subunits to mRNA)

(1/2 m each)

[2]

(iii) State **one** other function of the poly-A tail.

- The longer the poly-A tail the longer the time need for the tail to be shorten to a critical length for mRNA degradation to set in completely (allowing more time for translation)

- (Proteins will bind to the 3' poly-A tail and) facilitate mRNA transport through the nuclear pore, out of the nucleus.

- Poly-A tail may be recognized by proteins which binds and prevent degradation

[1]

(c) In mammals, sex is determined by the X and Y chromosomes, females being XX and males XY. In females, the expression of all the genes on one of the two X chromosomes in each cell is inactivated throughout the life of the female. This ensures that the effective dosages of products of X-linked genes are equal in males and females since a double dose of X-linked genes may potentially be toxic.

Suggest if the inactivation of gene expression on the X chromosome occurs via chromatin modification or at translation initiation. Explain your answer.

- Chromatin level
- *Link chromatin structure to gene expression*: Chromatin **condensation** → **long term repression/inhibition** of the genes
- repressing at chromatin level saves resources because cells do not need to use resources to produce the mRNA only to not use the mRNA for translation at all / idea of many mRNA would be formed from 1 gene , it is inefficient/ use of much more resources to inactivate so many mRNA compared to just inactivating 1 gene

[3]

[Total: 11]

4 Viruses infect both eukaryotic and prokaryotic cells.

Fig. 4.l is an electron micrograph of a eukaryotic cell infected by an enveloped virus such as the Zika virus.

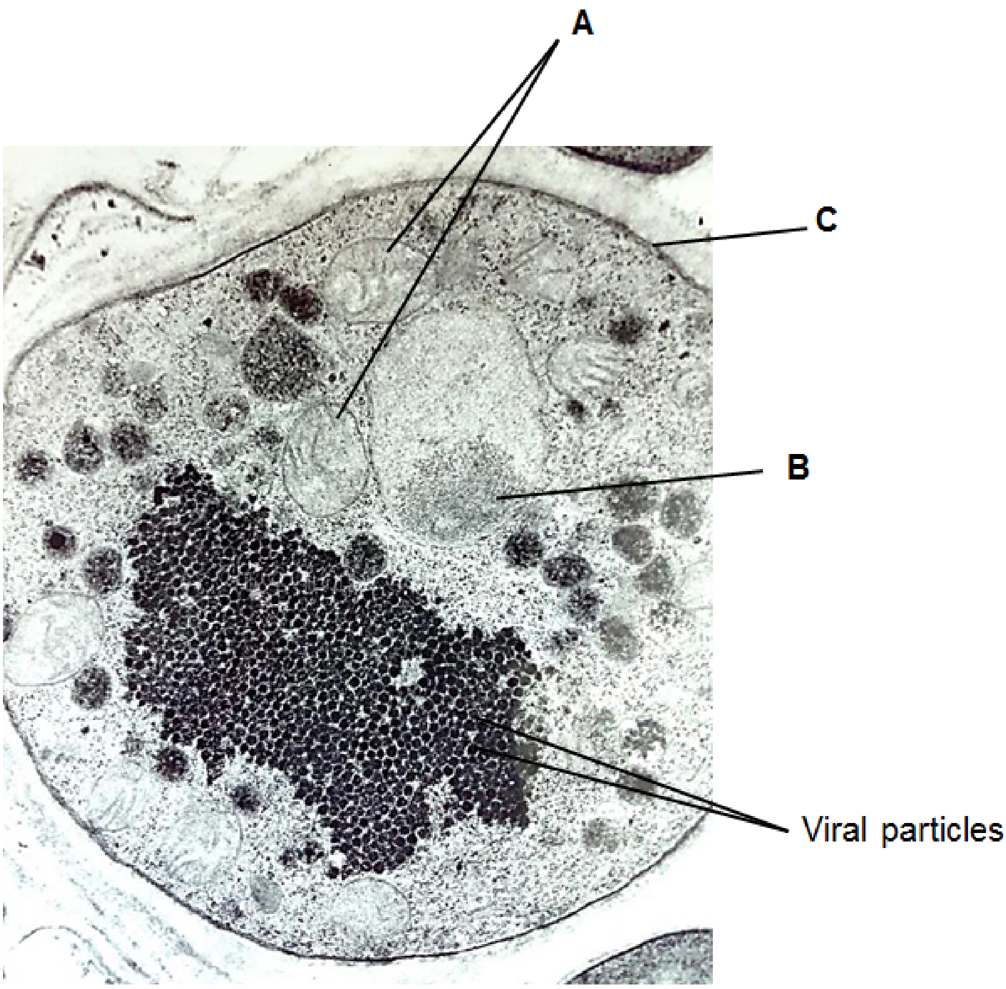


Fig. 4.1

(a) **(i)** Identify structures **A** and **B** visible in Fig. 4.1

A : mitochondria (reject mitochondrion)
B: Nucleolus (accept nucleus) [2]

(ii) Describe the role of structures **A** and **C** in the reproductive cycle of viruses such as the Zika virus.

- Mitochondria synthesizes **ATP** (reject release/ produce energy) during **aerobic respiration**
- For **activation of amino acids** during the **synthesis of viral proteins**
  Or Movement of **vesicles** containing **viral proteins/ surface glycoproteins** *reject capsid proteins* (to the surface membrane/ structure C/ between organelles) / AVP
- Structure C: cell surface membrane contains **specific receptors** to allow **viral surface proteins** to **attach/ bind** (via complementary shape) to the host cell (to enter / recognize specific host cell) OR viral particle **buds off** from the cell surface membrane, acquiring the **viral envelope** [3]

Bacteriophages are viruses that infect prokaryotic cells. The lambda bacteriophage is an example of a temperate bacteriophage that infects *Escherichia coli.*

Fig. 4.2 shows the changes in the number of extracellular lambda phages after they are introduced into a culture of *E.coli.*

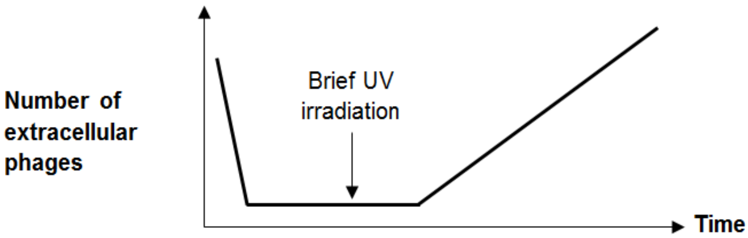


Fig. 4.2

(b) With reference to the reproductive cycle of temperate bacteriophages,

(i) explain why the number of phages increased after a brief UV irradiation.

- UV triggers switch from **lysogenic cycle** to lytic cycle / induction of lytic cycle/ causes prophage to be **excised** from host (bacterial) chromosomes
- Enzymes in host cells/ host cell machinery (give as least 2 examples) replicate viral DNA/ transcribe and translate the viral DNA into new viral components;
- **Lysis** of host cells to release bacteriophages;

[3]

(ii) explain why bacteriophage infection may be beneficial to bacteria population.

- A **small region of bacterial DNA** may be **excised** together with the prophage (and packaged)
- The resulting transducing phages **infect other / recipient bacteria** and newly infected cell **acquires the original bacterial DNA**
  Must have idea of DNA transfer
- **Transduction** increases **genetic recombination**/ increase **genetic variation** → increase adaptability of the bacteria to changes in environment / have selective advantage when the environment changes / give examples
  Reject allows bacteria to have selective advantage / advantageous traits w/o reference to gaining of advantageous genes

[3]

[Total:11]

5 Skin cancer cells may be grown in culture and examined using the technique of immunofluorescence in which antibodies are used to attach fluorescent dyes to specific molecules within the cells.

Fig. 5.1 is an immunofluorescent light micrograph of skin cancer cells. A particular type of protein is stained with the dye and appears as pale regions in the skin cancer cells.

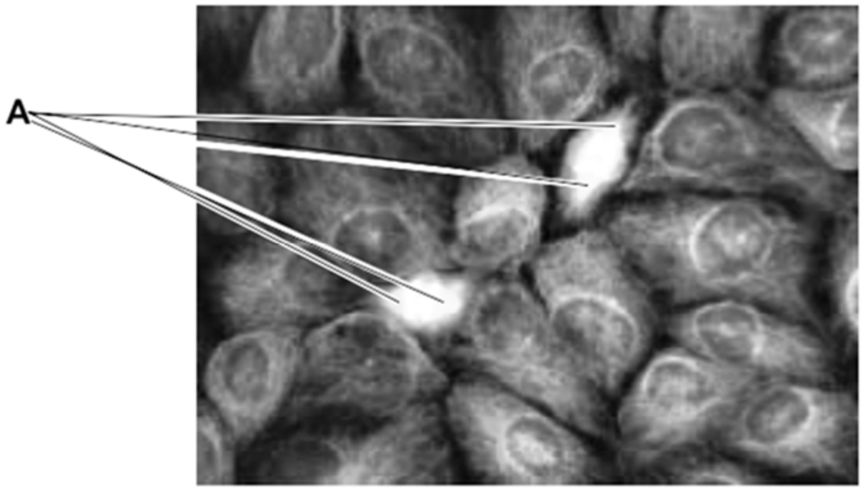


Fig. 5.1

(a) (i) Before the skin cancer cells could be stained with antibodies, the cells had to be fixed and treated with a mild detergent to increase the permeability of the cell surface membranes.

Explain the purpose of this step.

- Membrane has a **hydrophobic core**;
- Antibodies that is **polar** in nature are unable to pass thru the membrane.

Accept: too **large** to pass through the membrane [2]

(ii) There are two cells in the process of dividing. Each of these cells has two areas stained heavily, labelled **A** on Fig. 5.1.

Suggest the identity of these two areas and outline their functions in these cells.

- spindle apparatus / spindle fibres
  accept: spindle / microtubules / tubulin / centrioles / microtubule organising centres / MTOCs

- Attach to **centromere/kinetochore protein** of the chromosomes;
- Can **elongate/shorten** to **move chromosomes** during **mitosis**

(accept if students describe function of spindle fibre in prophase/anaphase);

Accept if centrioles given as identity
- forms poles of the cell ;
  organises the spindle ; [3]

(iii) Suggest why the proteins stained in the cytoplasm of the non-dividing cells in Fig. 5.1 are not evenly distributed.

- forming cytoskeleton/actin filaments in the cells.
- to maintain shape of cells.
- to help to support/anchor organelles in different parts of the cells. [1]

For movement of transport vesicles/secretory vesicles

(b) Explain **two** ways in which the behaviour of chromosomes in prophase of meiosis I differ from prophase of mitosis.

- During prophase I, homologous chromosomes **pair up**. However, in prophase of mitosis, homologous chromosome doses not pair up;

- During prophase I, chiasmata formed and **crossing over between non-homologous sister chromatids occurs**. However, in mitosis, no crossing over occurs. [2]

(c) Some chemicals known to inhibit the cell cycle are used as drugs for the treatment of cancer.

A particular drug was found to be most effective when applied to cancer cells in the G2 phase of the cell cycle.

Suggest the possible mechanism of this drug.

- Inhibit the condensation of chromosomes;
- Inhibit the replication of centrioles/centrosome;
- Inhibit the organisation of spindle fibres within the cells;
- Reject: "inhibit protein synthesis" unless specifically mention that inhibition of synthesis of proteins required for chromosome condensation;

Compulsory point:

- Induced a G2 check point and prevent the cells from entering mitosis; [3]

[Total: 11]

6 A castor oil plant, taken from a line which was known to be pure-breeding for the black and smooth seed coat, was crossed with a plant of unknown genotype.

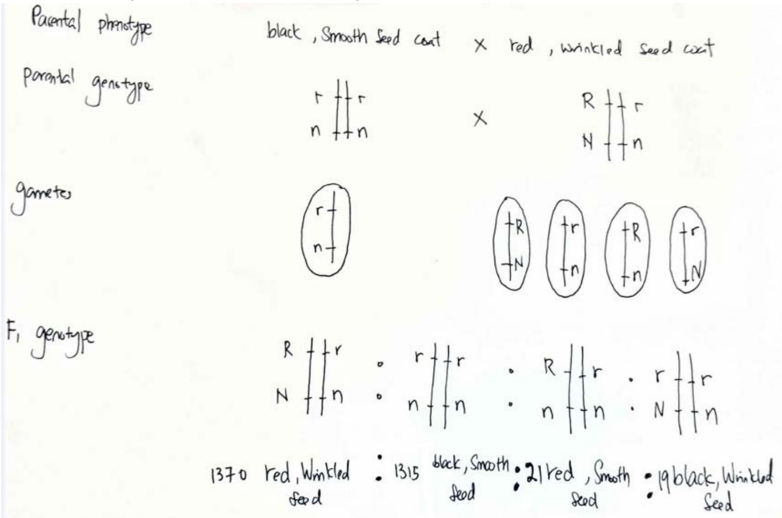
The cross gave the following F1 results.

black, smooth seed	1315
red, wrinkled seed	1370
red, smooth seed	21
black, wrinkled seed	19

(a) Draw a genetic diagram to explain these results.

Use the following symbols.

R red seed; **r** black seed; **N** wrinkled seed; **n** smooth seed.



[4]

(b) In another experiment involving two other characteristics of the castor plant, repeated test crosses with a plant which is heterozygous at both gene loci only produce progeny with 2 phenotypic classes instead of the expected 4.

Explain the above observation.

- The two genes are **<u>closely located</u>** on the same chromosome
- Chiasmata cannot be formed/crossing over cannot occur between the two gene loci.
- Always inherited together/completely-linked/tightly-linked
- Therefore, no formation recombination gametes/chromosomes. [3]

(c) In a different plant species, the type of seed coat is controlled by a single gene, B/b.

This gene encodes for an enzyme that converts the glucose formed during photosynthesis into starch for storage. Starchy seeds remain relatively smooth.

On the other hand, the homozygous recessive condition disrupts this conversion, producing seeds with high sugar content. When these seeds dry up, they shrivel and become wrinkled.

Explain why heterozygous plants for this gene, Bb, have the same phenotype as homozygous dominant plants, BB.

- Heterozygous plants possess the dominant allele B and can produce the enzyme that resulted in the conversion of glucose to starch. (these seed remain smooth)/ one copy of dominant allele can produce sufficient amount of functional enzyme;
- Recessive allele encodes for non-functional enzyme.
- The (single) dominant allele is able to mask the effect of the recessive allele.
- (Named phenotype) smooth seed (coat) [3]

[Total:10]

7

(a) Explain the part played by water in the production of ATP during photosynthesis.

- Photolysis of water;
- (water is) split using light energy;
- Provides electrons/hydrogen ions (H+ for ATP production);
- Replaces electrons lost by chlorophyll (a) in photosystem II;
- (is it repeat point?) Provides hydrogen ions for ATP production/ release oxygen;

½ mark each [2]

Most ATP is made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another.

Fig. 7.1 shows three such membrane systems.

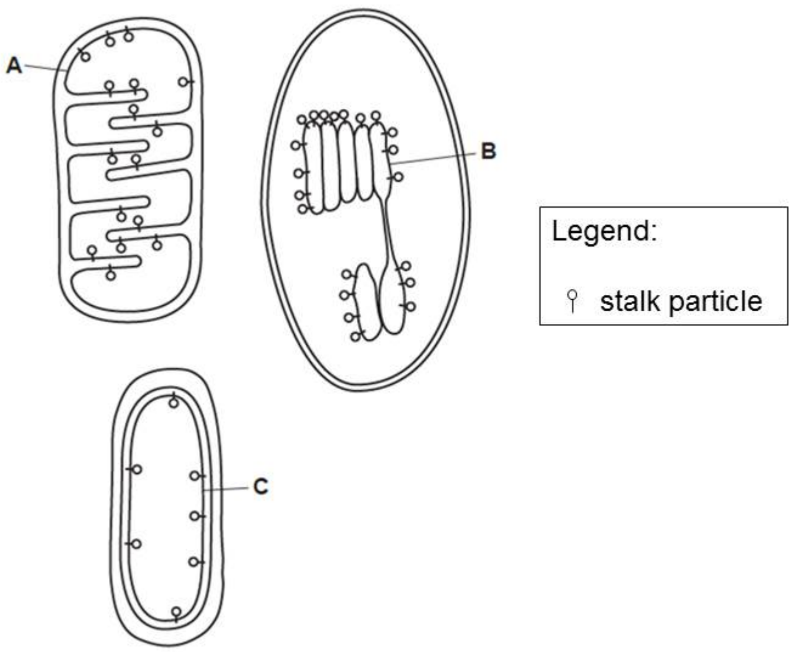


Fig. 7.1

(b) **(i)** Draw arrows onto each of the membrane systems in Fig. 7.1 to show the direction in which protons are pumped. [1]

Arrowhead in the opposite direction from the head of stalk particle;

(ii) Describe the role of membrane **B**. [3]

- Thylakoid membrane;
- Site for light-dependent reactions in photosynthesis;
- Increase surface area;
- To hold light harvest complex, photosystems (I &II), photosynthetic pigments, stalk particles/ ATP synthases;
- electron carriers in sequence from high energy to low energy;
- membrane is impermeable to H^+, H^+ is trap in thylakoid space in high concentration;

Fig. 7.2 summarises the reactions which occur in the Calvin cycle.

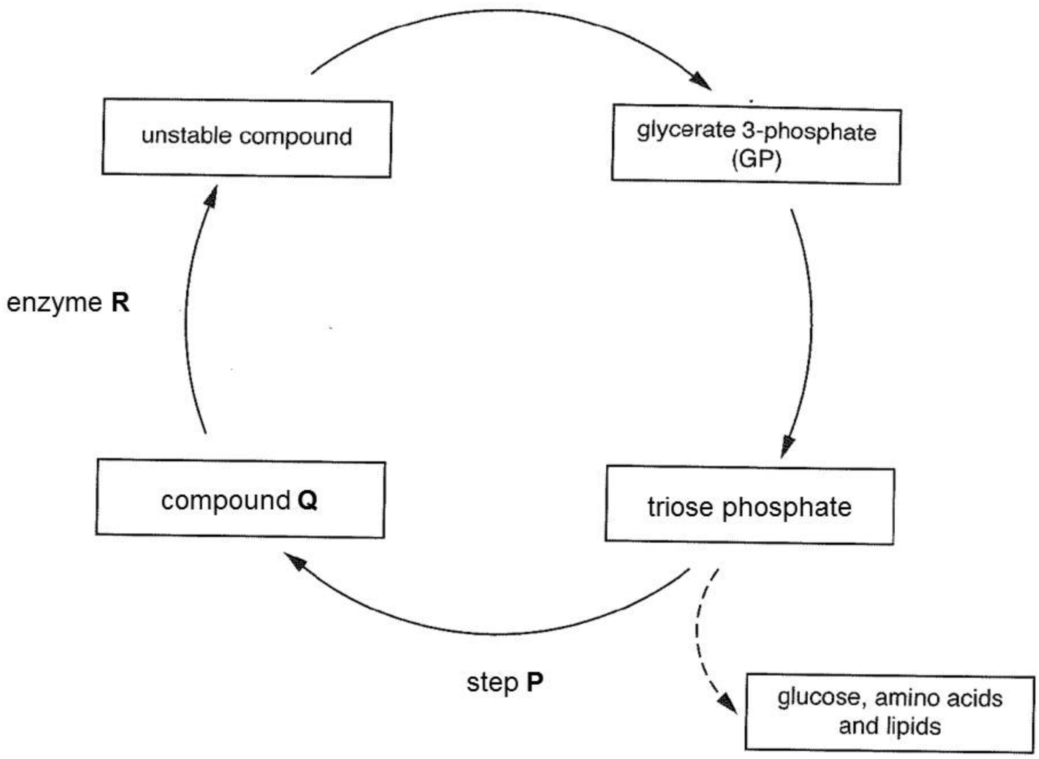


Fig. 7.2

(c) **(i)** Describe step **P**.

- RuBP regeneration;
- Use (3) ATP;
- (the carbon skeletons of) 5 molecules of G3P are rearranged into 3 molecules of RuBP; [2]

(ii) Some biologists describe enzyme **R** as 'the most important enzyme in our biosphere'.

Explain why they might hold this opinion.

- Identify R - rubisco / ribulose bisphosphate carboxylase (oxygenase);
- ref to carbon fixation/ first enzyme in Calvin cycle that combines CO_2 with RuBP to form glycerate 3-phosphate;
- glycerate 3-phosphate is used to make glucose;
- key role in carbon cycle ;
- only / main, route into food chains for carbon/ idea that glucose is used by plant itself for growth as well as by primary consumers and subsequent consumers (for respiration);

- the major route out of the atmosphere for carbon dioxide ; [3]

[Total:11]

8 Reproduction in seahorses, *Hippocampus*, is unusual as it is the male rather than the female that becomes pregnant. The male has a brood pouch located on its tail. The larger the male, the larger the pouch. The female transfers unfertilised eggs into the pouch. The larger the female the more eggs are produced that can be transferred to the brood pouch. The male releases sperm onto the eggs and they are fertilised. The male carries the developing brood for a period of several weeks until he finally gives birth.

Research into seahorse populations has revealed the following.

- They are monogamous. A male and female remain together for the whole mating season.
- Within a population, mates are selected by size. Large females mate with large males and small females mate with small males.
- Few intermediate sized individuals are produced and they have low survival rate.

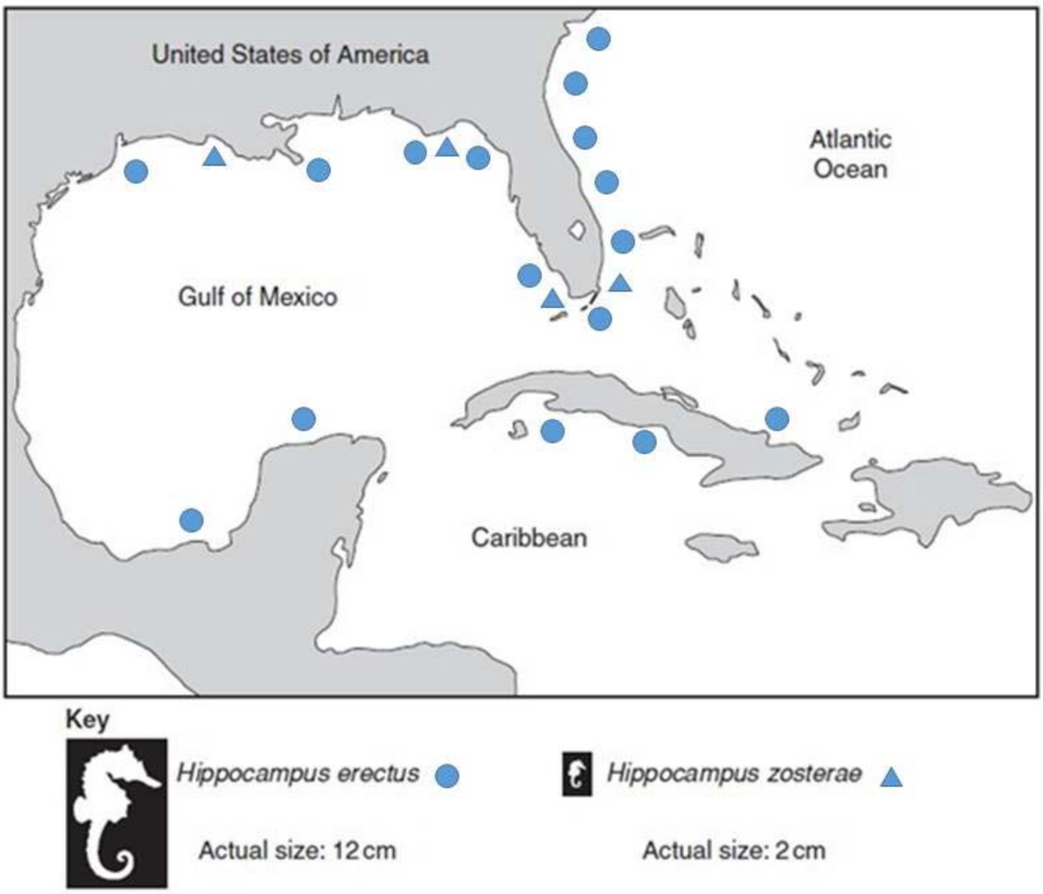


Fig. 8.1

Two different species of seahorse are found in the coastal region shown in Fig. 8.1. The ranges of these seahorse species overlap in many areas of these waters.

The two seahorse silhouettes are not drawn to scale.

(a) Using the information given, state the type of speciation that has occurred in the seahorses and explain your answer.

- Sympatric;
- ranges of two species, overlap/close together with correct ref to named area of map;
- no geographical barrier;
- ref to behavioural/genetic/physiological/prezygotic barrier;
- to prevent gene flow → resulting in reproductive isolation; [3]

(b) The type of natural selection that can produce the type of speciation that has occurred in seahorses is known as disruptive selection.

Explain how disruptive selection occurs in seahorse populations.

- ref to mate selection by size;
- ie large with large or small with small ref to monogamy;
- ref to intermediate sizes, at disadvantage/selected against/ora;
- intermediate do not survive and reproduce to pass on alleles/ora;
- suggested reason why intermediate at disadvantage/ora max 3 [3]

(c) Fig. 8.2 shows the phylogenetic tree of three closely related species of seahorses based on nucleotide sequences, with ages estimated from fossils and biogeographical data.

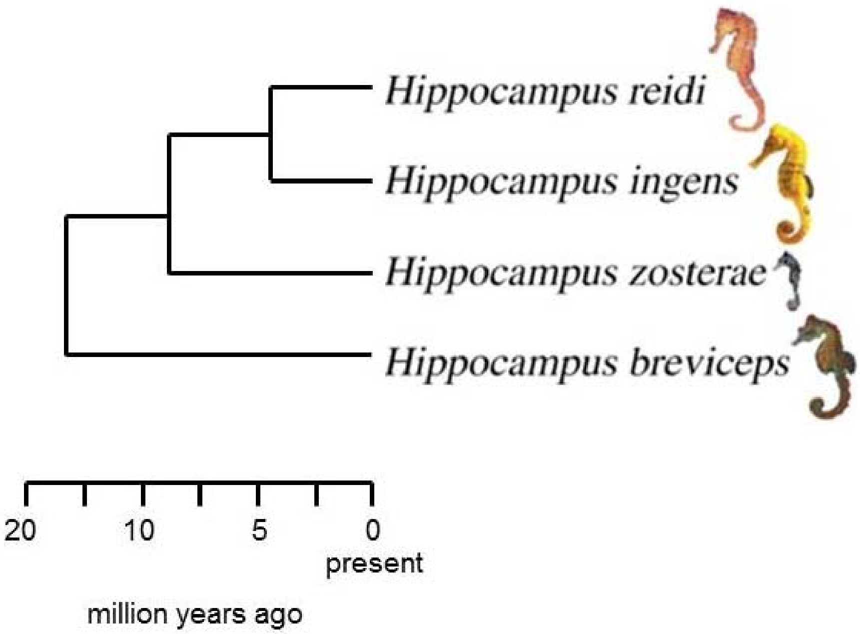


Fig. 8.2

(c) Describe the advantages of using nucleotide sequences in reconstructing the phylogenetic relationships.

- Comparing **molecular evidence** like nucleotide sequences was **unambiguous** and **objective,** was **quantifiable**, thus can be used to measure the **degree of relatedness** between different organisms. and was open to **statistical analysis.**
  In contrast, classification based on **observable** characteristics like anatomy may be **subjective**, since it is not easy to determine whether similar structures arise due to a **recent common ancestry** or due to **convergent evolution.**
- Molecular methods are able to **differentiate two organisms with similar morphologies based on molecular differences**
- Molecular methods also reveal that some **major phenotypic differences** may actually be **due to small genetic differences**.

- It can be used to establish the **molecular clock** to estimate organisms' **evolutionary history**/**phylogeny**, and detect **silent genes**.
- Based on the molecular clock, the speciation event in divergent evolution to form ***Hippocampus zosterae,*** and the ancestor of ***Hippocampus reidi*** and ***Hippocampus ingens***, was determined to occur at 5 million years ago.
- Organisms with more molecular similarities like ***Hippocampus reidi*** and ***Hippocampus ingens*** are grouped more closely together, and thus determined to be more closely related [3]

[Total:9]

9 Fig. 9.1 shows the molecular structure of an antibody.

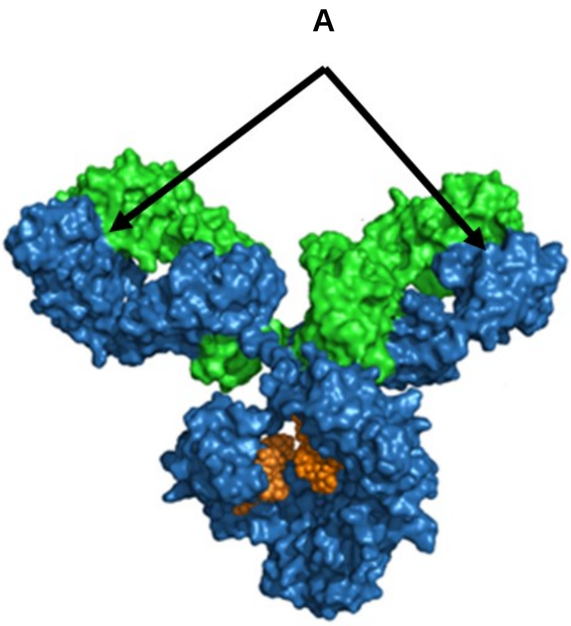


Fig. 9.1

(a) Explain how diversity at regions labelled **A** is generated in developing antibody-secreting cells.

- There are multiple **gene segments** at heavy and light chain gene loci;
- **Somatic recombination** occurs during B cell maturation and development, where there is **DNA rearrangement** to assemble gene segments:
- At **heavy chain** gene locus, one **V** gene segment, one **D** gene segment and one **J** gene segment are rearranged together to form **VDJ** exon to code for **variable domain** of the heavy chain;
- At **light chain** gene locus, one **V** gene segment and one **J** gene segment are rearranged together to form **VJ** exon to code for **variable domain** of the light chain;
- There is **combinatorial diversity**, where different combinations of light chain and heavy chain variable domains lead to **different antigen – binding sites which bind different antigens**, generating different antibodies of **different antigen specificities**; [2]

Fig. 9.2 shows the typical antibody concentration in the serum of a patient during a primary and a secondary immune response to the same antigen.

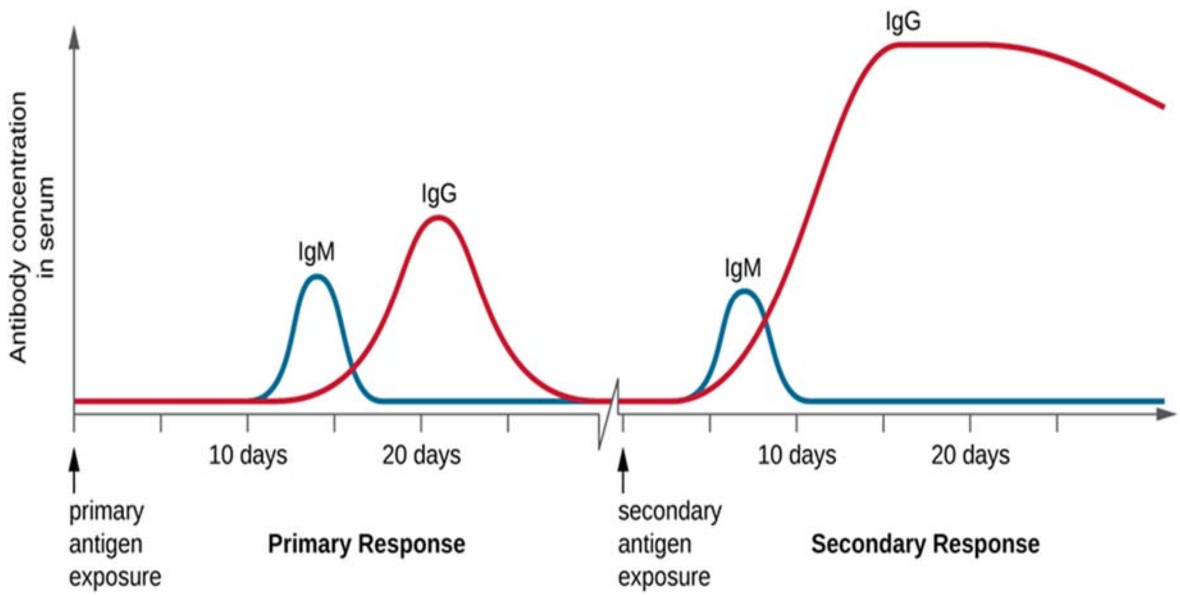


Fig. 9.2

(b) **(i)** State **two** significant differences between the primary and secondary immune responses.

Primary immune response	**Secondary immune response**
Antibody secretion start 10 days after antigen exposure	Faster secretion of antibody 5 days after antigen exposure
Ref. to the idea lower amount to Ig G (as compared to secondary immune response)	higher amount of Ig G (as compared to primary immune response

[2]

(ii) Explain the differences stated in **(a)**.

Slower primary response and faster secondary response:

- Antigen – presenting cells (APCs) to engulf and **present antigen** to CD4 T cells to activate these T cells to divide and differentiate to produce T helper cells and memory T helper cells;
- Antigen – specific B cells to take in antigen via antigen binding to the B cell receptor (cell surface membrane immunoglobulin, IgD and IgM, with lower affinity to antigen as compared to during secondary immune response) and subsequent **receptor-mediated endocytosis**;
- Activated B cells to **divide** and **differentiate** to form more plasma cells and memory B cells;
- Antigen – specific B cells increased in numbers as there is **clonal expansion** of B cells during primary immune response;
- Immunoglobulins in memory B cells have higher affinity to antigen (refer to Fig. 5.3) due to B cell going through **somatic hypermutation** during cell division and the consequent **affinity maturation**;
- Memory B cells express higher levels of MHC class II molecules on their cell surface membranes, which facilitate antigen presentation; [3]

- Increased expression of co-stimulatory molecules on the B cell surface membrane;

- Memory B cells can initiate critical interactions with helper T cells at lower antigen concentration than naïve B cells, differentiation and antibody production start earlier after antigen stimulation than in the primary response;

  [max 2m]

Greater amount of IgG:
- Earlier generation of plasma cells from primary immune response, immediately secrete IgG antibodies;
- B cells has undergone **class switching**, which produces more B cells which secrete IgG;

[max 1m]

Fig. 9.3 shows the production of all blood cells from Cell **B**.

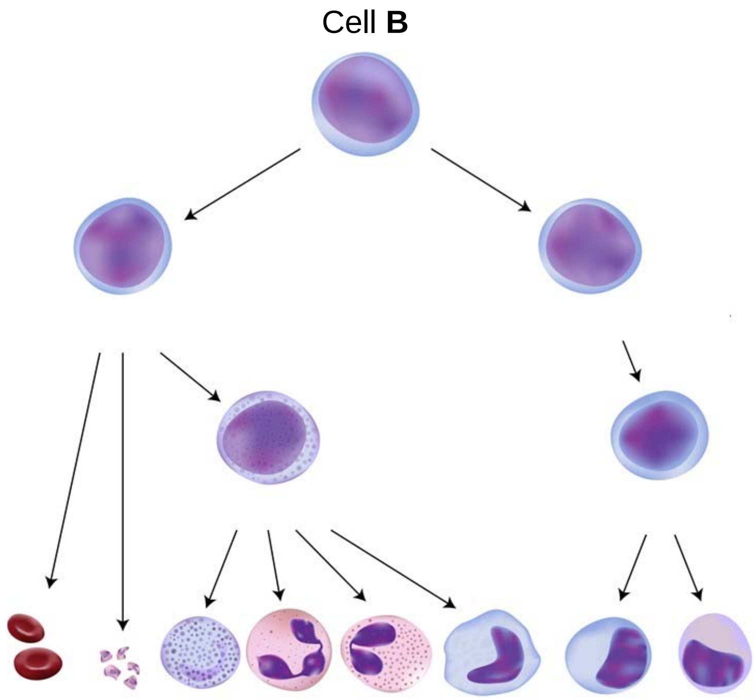


Fig. 9.3

(c) Identify Cell **B** and explain why, as shown in Fig. 9.3, it must have the characteristics of a stem cell.

- Cell B: haematopoietic stem cell / blood stem cell;
- Haematopoietic stem cell is able to **self-renew via mitosis**, to give rise to more cells which can replace worn-out / damaged blood cells;
- Haematopoietic stem cells are **multipotent**, being able to differentiate into various specialised blood cells;
- Haematopoietic stem cells are **unspecialised** with no specialised cell structure and function;

[4]

[Total: 11]

10 An analysis of ice cores from the Artic and Antarctic can provide information about the composition of the Earth's atmosphere over thousands of years.

Fig. 10.1 shows the concentrations of carbon dioxide and methane measured in ice cores, dated between 1000 and 2000AD.

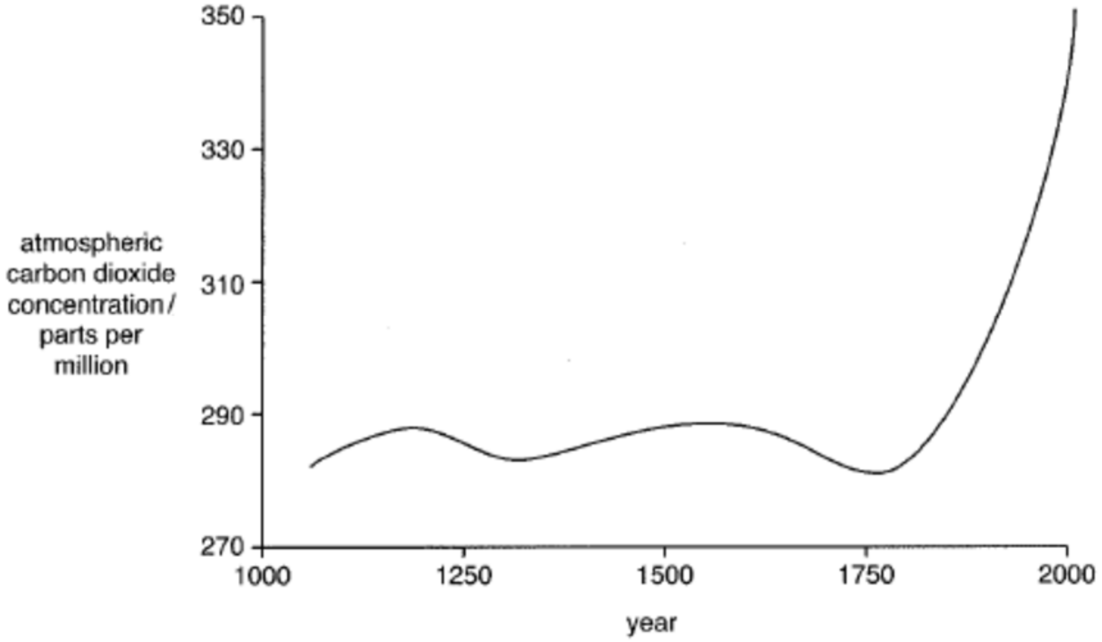


Fig.10.1

(a) Describe and explain the data in Fig. 10.1 from 1750AD onwards.

- (Describe data trend and quote) Atmospheric carbon dioxide concentration **increase sharply** from 280 to 350 parts per million from 1750 to 2000AD;
- (explain) Rise of both atmospheric carbon dioxide and methane concentration after 1750 is due to industrial revolution, rise in human population, rise in human activities, rise in demand of energy consumption, deforestation for other land uses and meat consumption; [2]

(b) Explain how increasing concentrations of gases, such as carbon dioxide and methane, are thought to cause global warming.

- Blanket of carbon dioxide and methane around the Earth;
- Allows high energy/ short wavelength rays from the Sun to enter Earth's atmosphere;
- When these are reflected from the surface of the Earth;
- They lose energy;
- And the longer wavelength/ lower energy/ infra-red/ heat rays cannot escape through the carbon dioxide blanket;(causes global warming) [2]

(c) As part of the Alliance of Small Island States (AOSIS), Singapore is considered a low-lying small island country. Like many other small island states, there are many fringing and patch coral reefs found around the smaller islands, south of Singapore mainland.

Describe how global warming can impact small island states like Singapore.

Rise in sea level→leading to loss of land

1 Melting of ice sheets results in extra water entering the ocean, leading to **increase in sea level.**

2 Melting of sea ice can lead to a positive feedback loop which **further accelerates the melting of polar ice caps.**

3 As oceans become warmer, the **water also expands, increasing the volume of the ocean water and leading to further rise in sea level.**

4 **Low-lying islands** / **coastal** regions may become **submerged**.

[Any 1 pair]

Death of coral reefs

5 Heat stress can cause **coral bleaching** / coral polyps expel the zooxanthellae.

6 This is because at higher temperatures, **zooxanthellae photosynthesis is disrupted** / zooxanthellae produces more toxic compounds

7 If temperatures remain above the bleaching threshold for prolonged periods of time, **corals will eventually die** from starvation and disease.

8 **Ocean acidification** affects hard corals as they cannot absorb the calcium carbonate to maintain their skeletons (and the stony skeletons that support corals will dissolve).

15 impact of loss of coral reef→ loss of (25% of) marine life [4]

[Total:8]